

Generative Models for Images

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THE OHIO STATE UNIVERSITY
COLLEGE OF ENGINEERING

Agenda

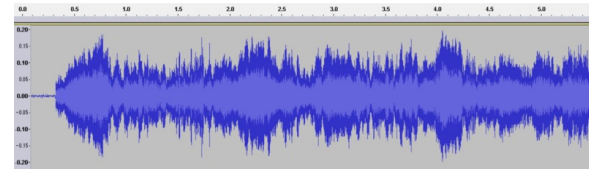
1. What are generative models? Examples?
2. Variational Auto-Encoder (VAE)
3. How to evaluate generative models for images?

Introduction

Challenge: understand and generate complex, unstructured inputs



Computer Vision



Computational Speech



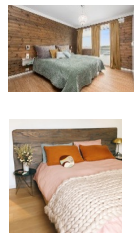
Natural Language Processing



Robotics



Generative models are learned from data



...



Data
(e.g., images of bedrooms)

Prior Knowledge
(e.g., physics, materials, ..)

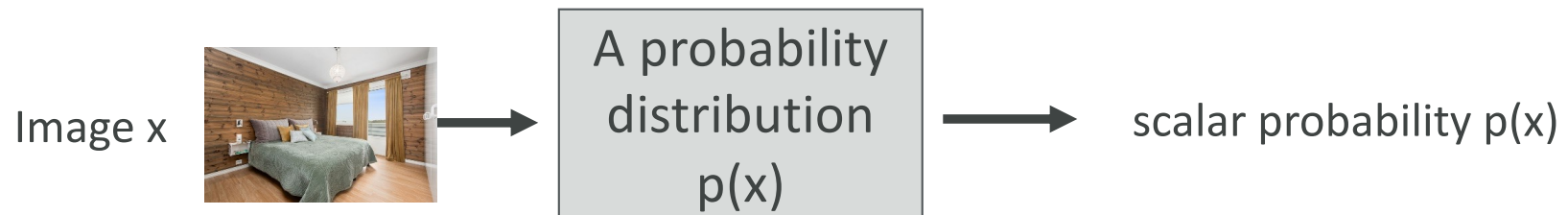
Prior knowledge always necessary, but there is a spectrum



Statistical View of Generative Models

Statistically, a generative model learns a **probability distribution** $p(x)$

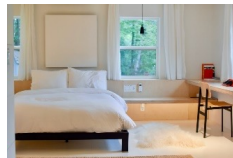
- **Data:** samples (e.g., images of bedrooms)
- **Prior knowledge:** parametric form (e.g., Gaussian?), loss function, optimization algorithm, etc.

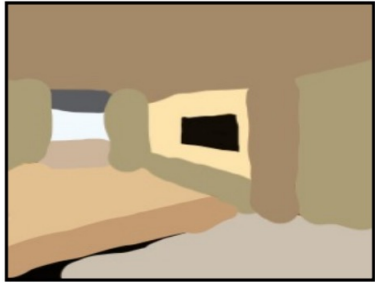


It is generative because **sampling from $p(x)$ generates new images**



...





Generative model
of realistic images



Stroke paintings to realistic images
[Meng, He, Song, et al., ICLR 2022]

“Ace of Pentacles”



Generative model
of paintings

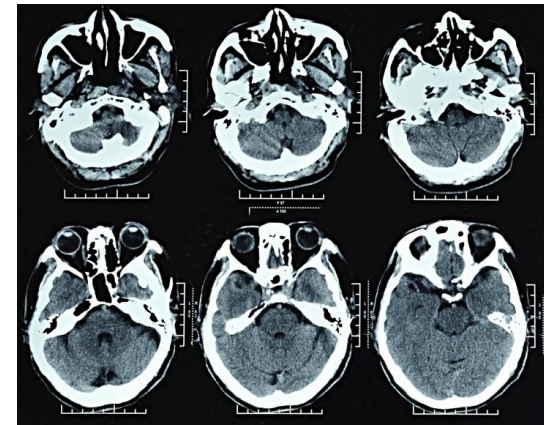


Language-guided artwork creation
<https://chainbreakers.kath.io> @RiversHaveWings



Generative model
of medical images

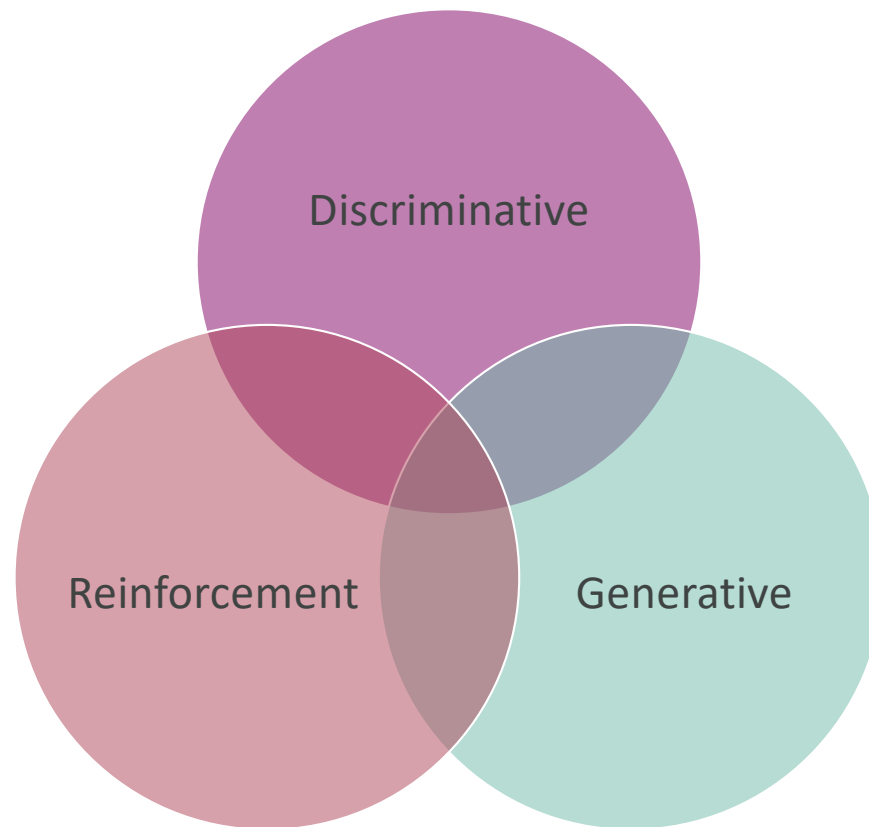
Generate



Medical image reconstruction

[Song et al., ICLR 2022]

Category of ML Problems



Discriminative vs. generative

Discriminative: classify bedroom vs. dining room



The image X is given. **Goal:** some decision boundary

- Requires **conditional distribution over label Y : $p(Y|X)$**
- E.g.: logistic regression, convolutional neural net, etc.

Generative: generate X

$Y=\text{Dining}$, $X=$



$Y=\text{Bedroom}$, $X=$



The input X is **not** given. **Goal:** generate X based on label

- Requires a model of the **joint distribution over both X and Y : $p(Y, X)$**

Discriminative vs. generative

Joint and conditional are related via **Bayes Rule**:

$$\begin{array}{c}
 \text{Conditional} \\
 P(Y = \text{Bedroom} \mid X = \text{img}) = \frac{P(Y = \text{Bedroom}, X = \text{img})}{P(X = \text{img})} \\
 \text{Joint} \qquad \qquad \qquad \text{Marginal}
 \end{array}$$

Discriminative: no need to model: $P(X = \text{img})$

Therefore, it cannot handle missing data: $P(Y = \text{Bedroom} \mid X = \text{img})$???

Images and Text

TEXT PROMPT an armchair in the shape of an avocado. . . .

AI-GENERATED
IMAGES



[Edit prompt or view more images](#)↓

$P(\text{image} \mid \text{caption})$

TEXT PROMPT a store front that has the word 'openai' written on it. . . .

AI-GENERATED
IMAGES



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Text2Image Diffusion Models

User input:

An astronaut riding a horse



Text2Image Diffusion Models

User input:

A perfect Italian meal



Text2Image Diffusion Models

User input:

泰迪熊穿着戏服，站在太和殿前唱京剧

A teddy bear, wearing a costume, is standing in front of the Hall of Supreme Harmony and singing Beijing opera



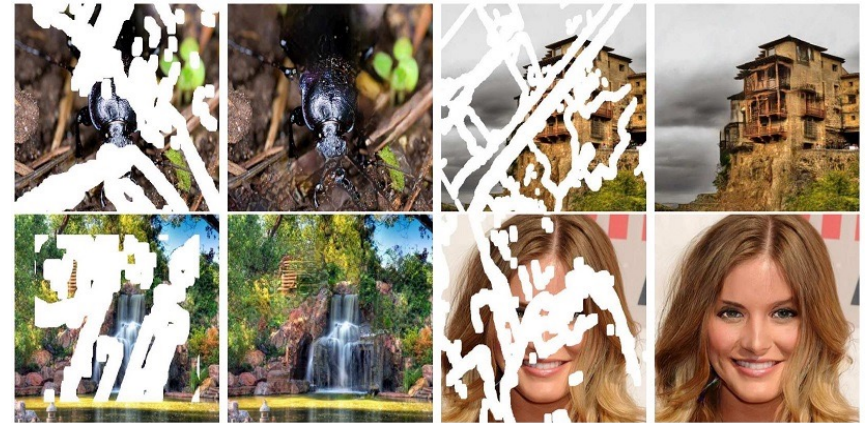
Example: Image Inverse Problems

$P(\text{high resolution} \mid \text{low resolution})$



Menon et al, 2020

$P(\text{full image} \mid \text{mask})$



Liu al, 2018

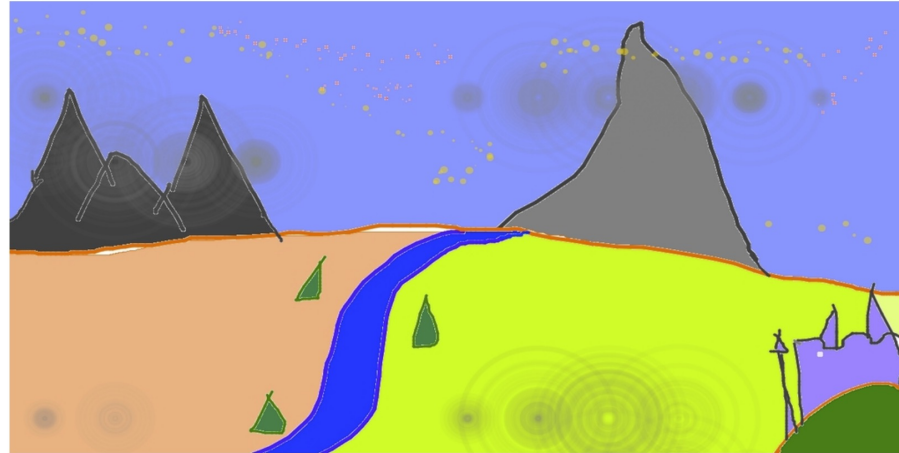
$P(\text{color image} \mid \text{greyscale})$



Antic, 2020

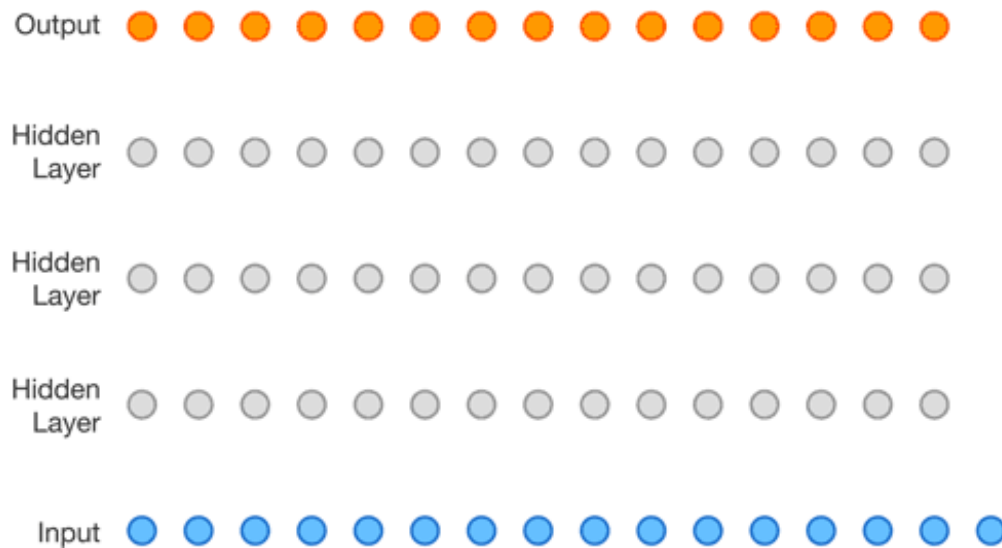
Example: Image Inverse Problems

User input:



WaveNet

Generative model of speech signals



Text to Speech



Parametric



Concatenative



WaveNet



Unconditional

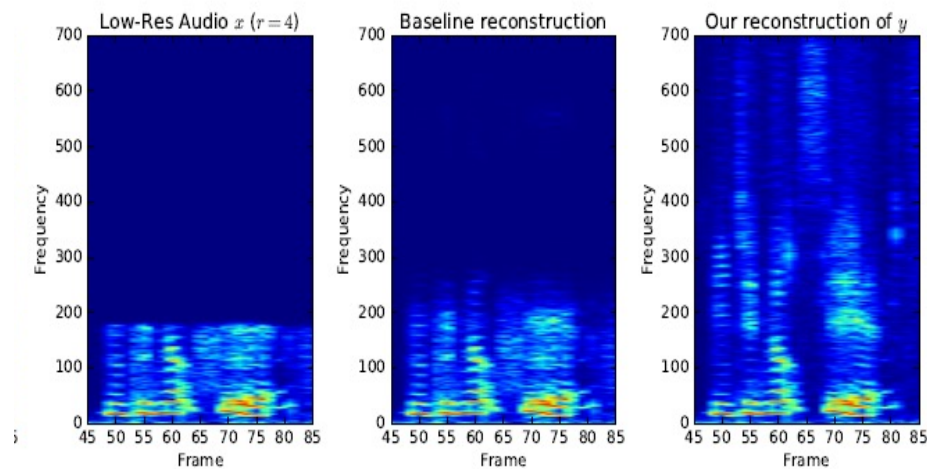


Music

van den Oord et al, 2016c

Audio Super Resolution

Conditional generative model $P(\text{high-res signal} \mid \text{low-res audio signal})$



Low res signal



High res audio signal

Kuleshov et al., 2017

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Video Generation

Suddenly, the walls of the embankment broke and there was a huge flood



Video Generation

a couple sledding down a snowy hill on a tire
roman chariot style



Generation for anomaly detection



High
probability
→



Generative model
of traffic signs



Outlier detection
[Song et al., ICLR 2018]

Recap: Generative Models vs. Traditional Stat

Traditional Statistics

- Usually assume simple models (e.g., Gaussian)
- Focus on distribution/parameter learning (e.g., Maximum Likelihood)
- Hard to adapt to complex data (e.g., images, audio, video)

Generative Models

- Adapted to real-world data
- Focus on the sampling task, not learning the distribution

Recap: Generative vs. Discriminative Models

Discriminative Models

- Restricted to certain tasks (i.e., classification & regression)
- Hard to handle missing/imbalanced data

Generative Models

- Helpful for imperfect data (e.g., missing)
- Closer to the meaning of “understanding” data

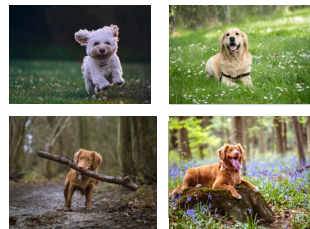
Getting into model details...

- Let's take discriminative models as an example...
- What are the components for ML discriminative models?
- 1. Representation:** Need to model the mapping from $x \rightarrow y$
 - Model architectures: SVM/XGBoost? RNN/CNN? Transformer?
- 2. Learning:** Need a **training** algorithm to learn the model from data (e.g., loss function)
 - Loss for discriminative models: Cross-ent, Euclidean distance...
 - Algorithm: GD, SGD, mini-batches...

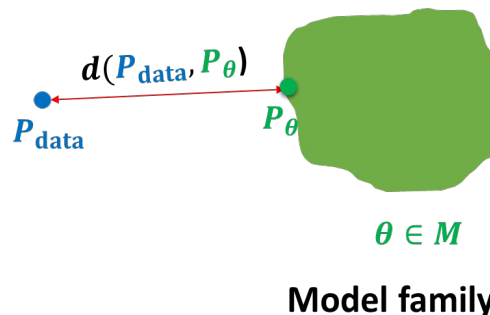
Challenges from generative models

Key: no longer predictions, but rather **distributions**!

1. **Representation:** How to model the **marginal/joint distribution** of complex data?
2. **Learning:** What is the right way to **compare distributions**?



$$\begin{aligned} \mathbf{x}_i &\sim P_{\text{data}} \\ i &= 1, 2, \dots, n \end{aligned}$$



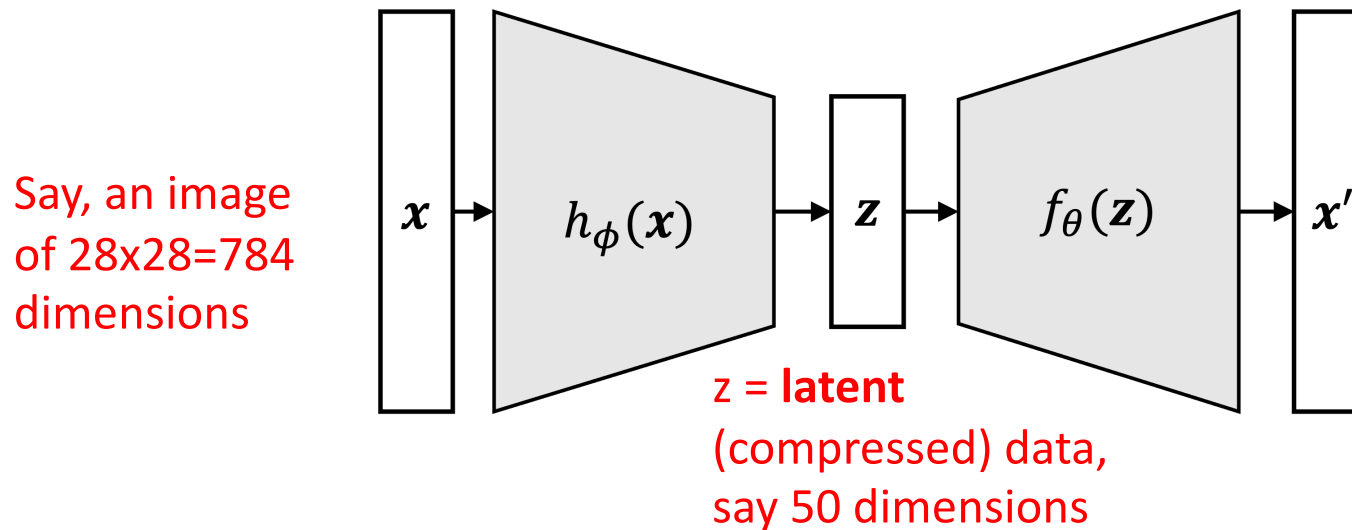
- **Other challenges:** How to obtain a sample? How to evaluate outcome?

Agenda

1. What are generative models? Examples?
- 2. Variational Auto-Encoder (VAE)**
3. How to evaluate generative models for images?

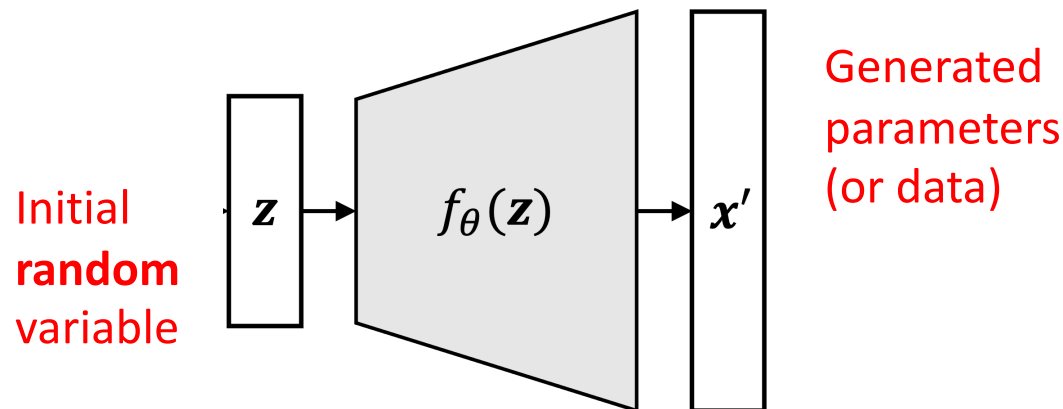
Variational Auto-Encoder (VAE)

- **Auto-encoder** Structure: **Encoder** network h_ϕ + **Decoder** network f_θ
- **Variational**: noise added at latent, for generation

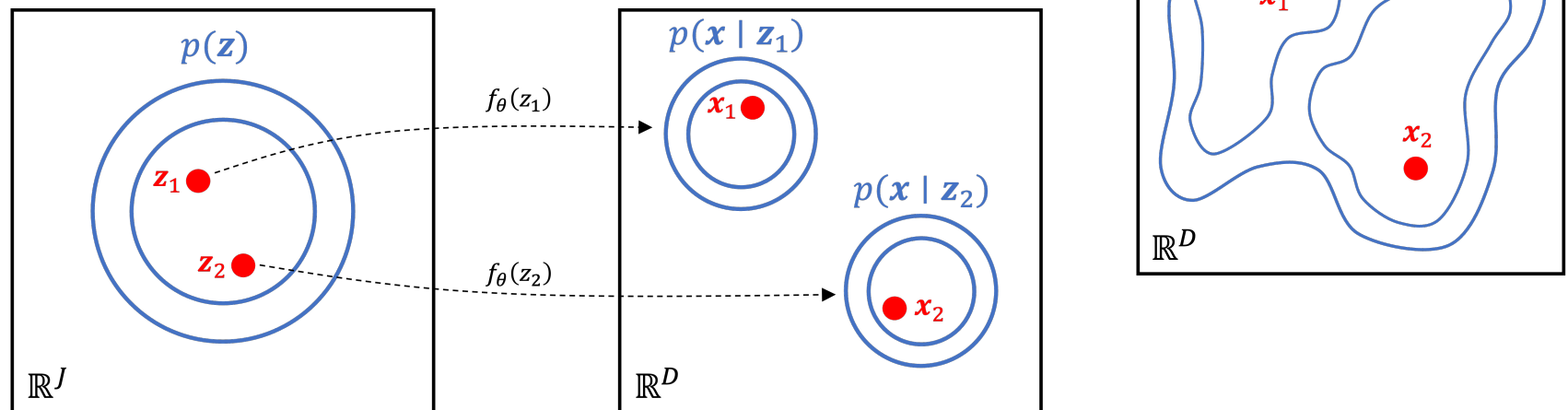


How to sample from VAE?

1. Generate a **random** variable $z \sim \mathcal{N}(\mu, \Sigma)$
2. One forward-pass of the **decoder** network $x' = f_{\theta}(z)$
3. (Optional): Re-sample again using x' (say, from a localized Gaussian)



VAE Sampling

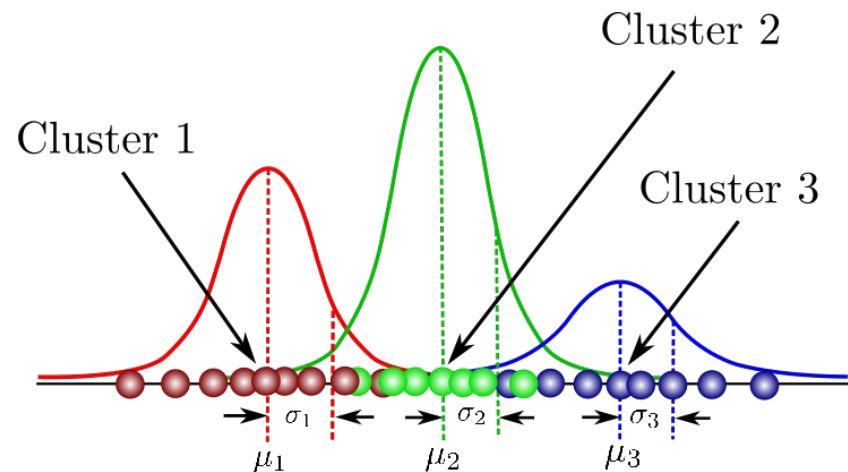


VAE: Distribution Representation (cont.)

- By law of total probability:

$$p_{\theta}(x) = \int p_{\theta}(x, z) dz = \int \underbrace{p_{\theta}(x|z)}_{\text{Decoder (Gaussian)}} \underbrace{p_{\theta}(z)}_{\text{Latent Dist (also Gaussian)}} dz$$

- This is **Gaussian mixture**, which is very complex



Learning VAE models

- Suppose we are given N training data (say, images): $x_1, \dots, x_N$
- Training goal: maximize (log-)likelihood:

$$\log p_{\theta}(x_1, \dots, x_N) = \sum_i \log p_{\theta}(x_i)$$

VAE: How to learn? (cont.)

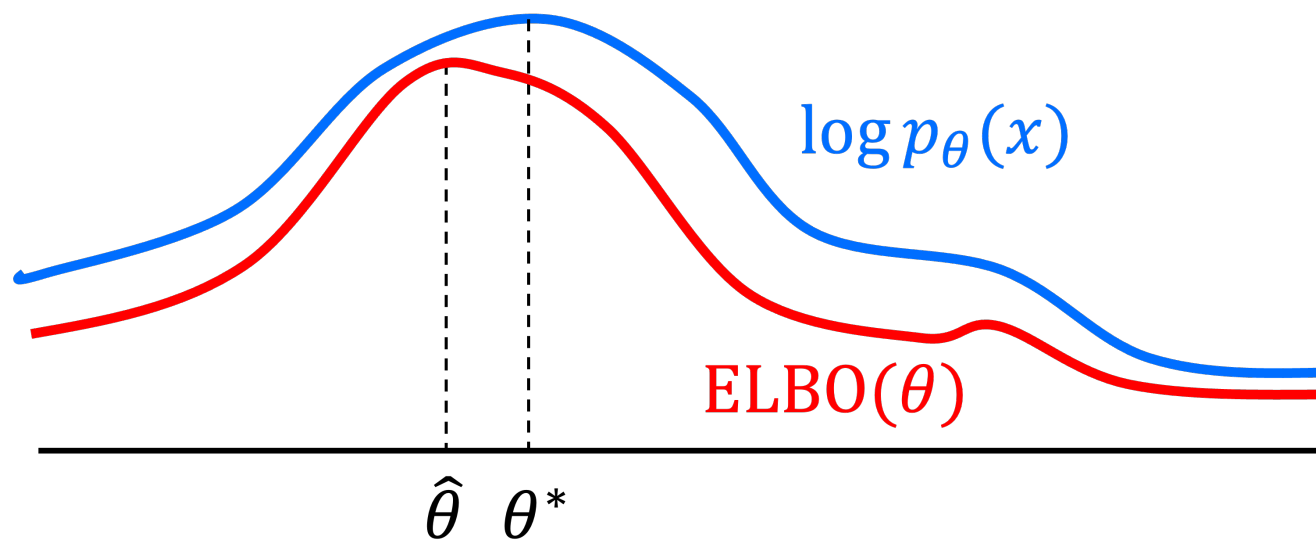
- Let's do some math...

$$\begin{aligned}\log p_{\theta}(x) &= \log \left(\int p_{\theta}(x, z) dz \right) && \text{(Law of total probability)} \\ &= \log \left(\int \frac{p_{\theta}(x, z)}{q_{\phi}(z|x)} q_{\phi}(z|x) dz \right) && \text{(Multiply and divide the same quantity)} \\ &\geq \underbrace{\int \log \frac{p_{\theta}(x, z)}{q_{\phi}(z|x)} q_{\phi}(z|x) dz}_{\text{ELBO}} && \text{(Jensen's inequality)}\end{aligned}$$

p_{θ} = Decoder/Generator
 q_{ϕ} = Encoder

- The last line is the **Evidence Lower BOund (ELBO)**

ELBO visualized



Optimization Algorithm

- Step 1: Obtain N datapoints (possibly from a random minibatch)
- Step 2: Compute the ELBO
$$ELBO = \int (\log p_{\theta}(x, z) - \log q_{\phi}(z|x)) q_{\phi}(z|x) d z$$
- Step 3: Calculate the gradient w.r.t. θ and ϕ
- Step 4: Gradient Update
- Iterate Steps 2-4 until convergence...

Optimizing VAE ELBO (cont.)

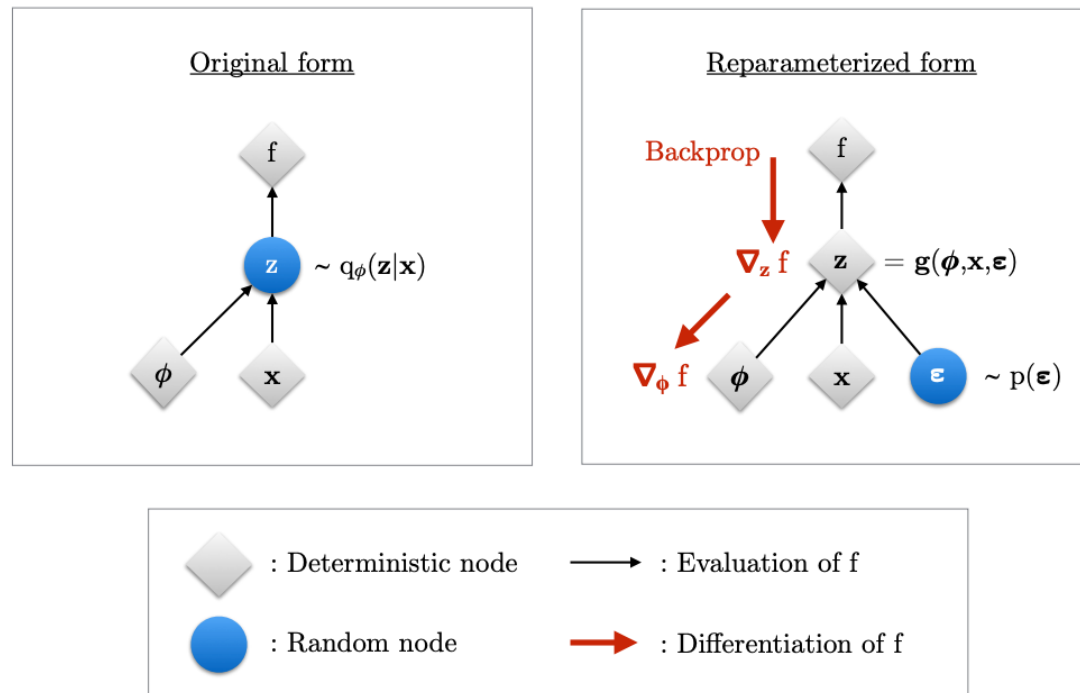
$$ELBO = \int (\log p_{\theta}(x, z) - \log q_{\phi}(z|x)) q_{\phi}(z|x) dz$$

- We need the those derivatives for gradient descent (w.r.t., θ and ϕ)
- It's easy to take derivative w.r.t. θ :

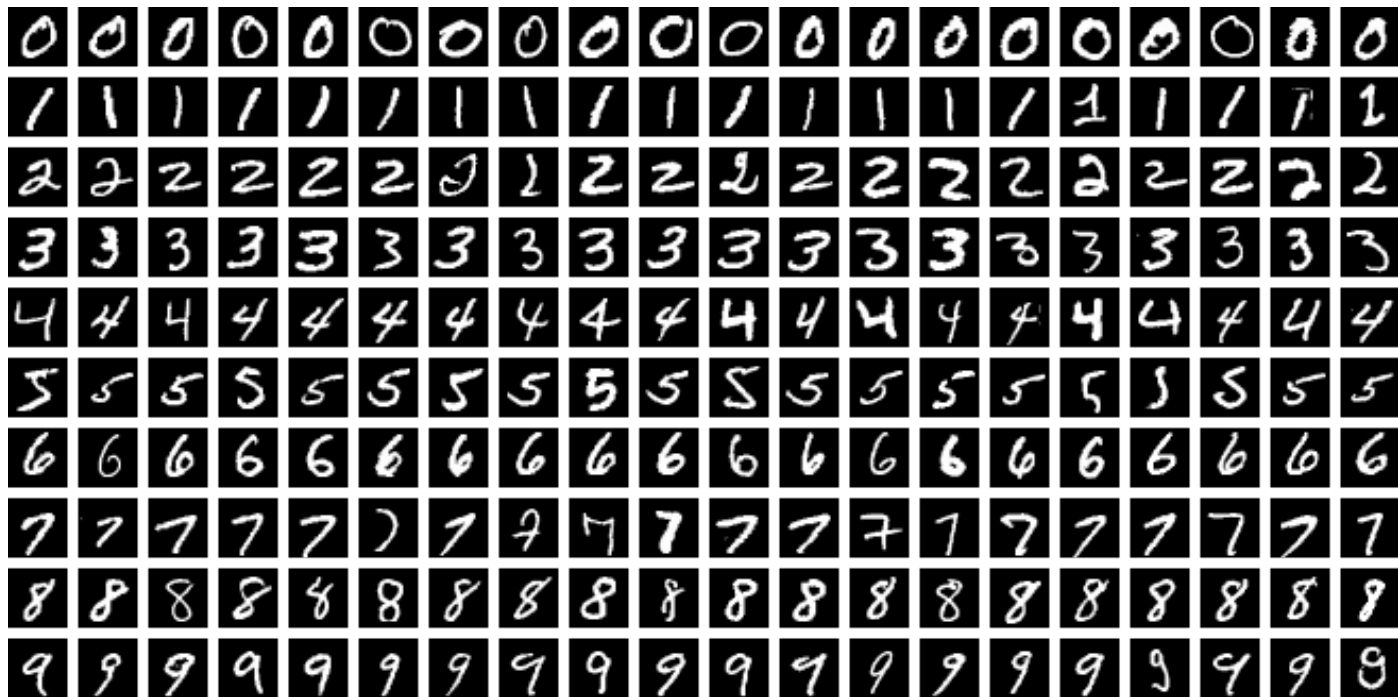
$$\frac{d}{d\theta} ELBO = \int \left(\frac{d}{d\theta} \log p_{\theta}(x, z) \right) q_{\phi}(z|x) dz$$

- It's very hard to take derivative w.r.t. ϕ ! There is an integral!
- How to solve for this? **Reparameterization trick**

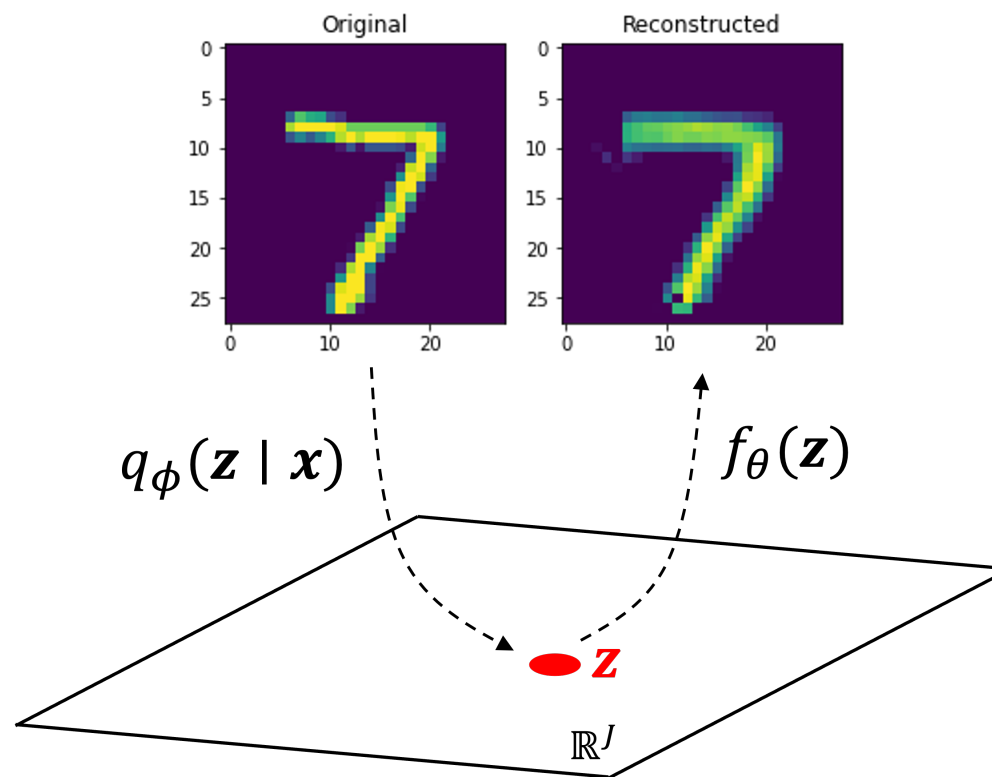
Reparameterization Trick



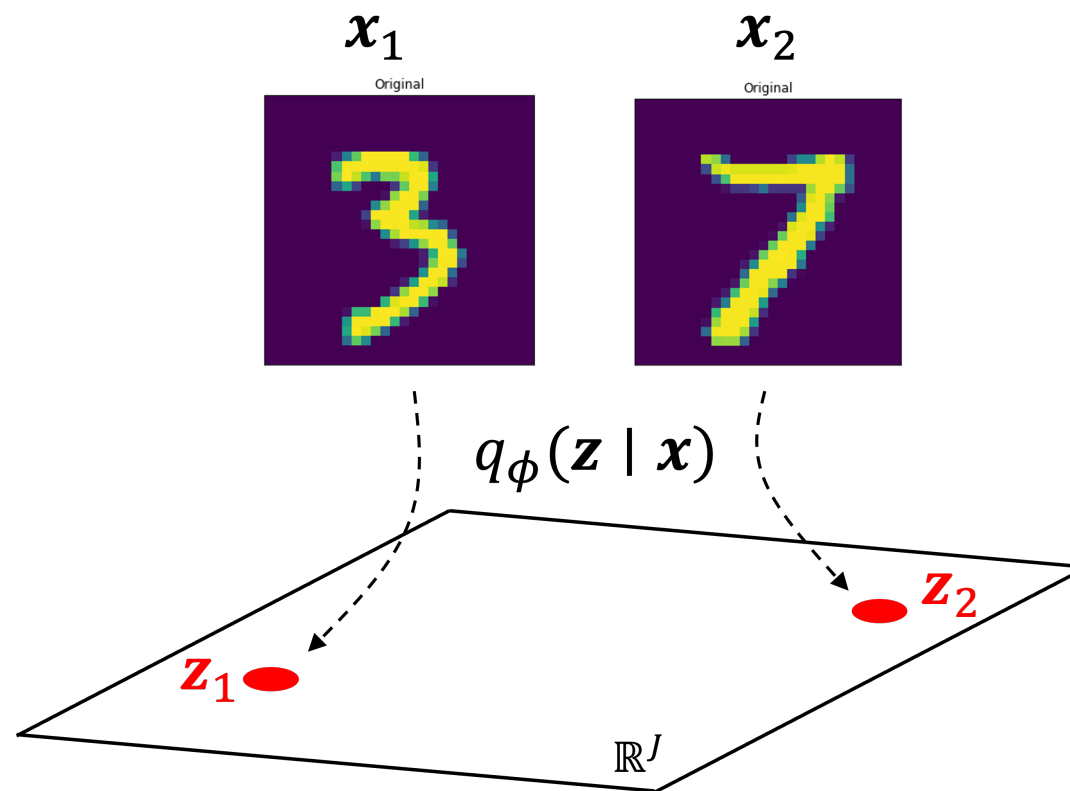
Applying VAE on MNIST



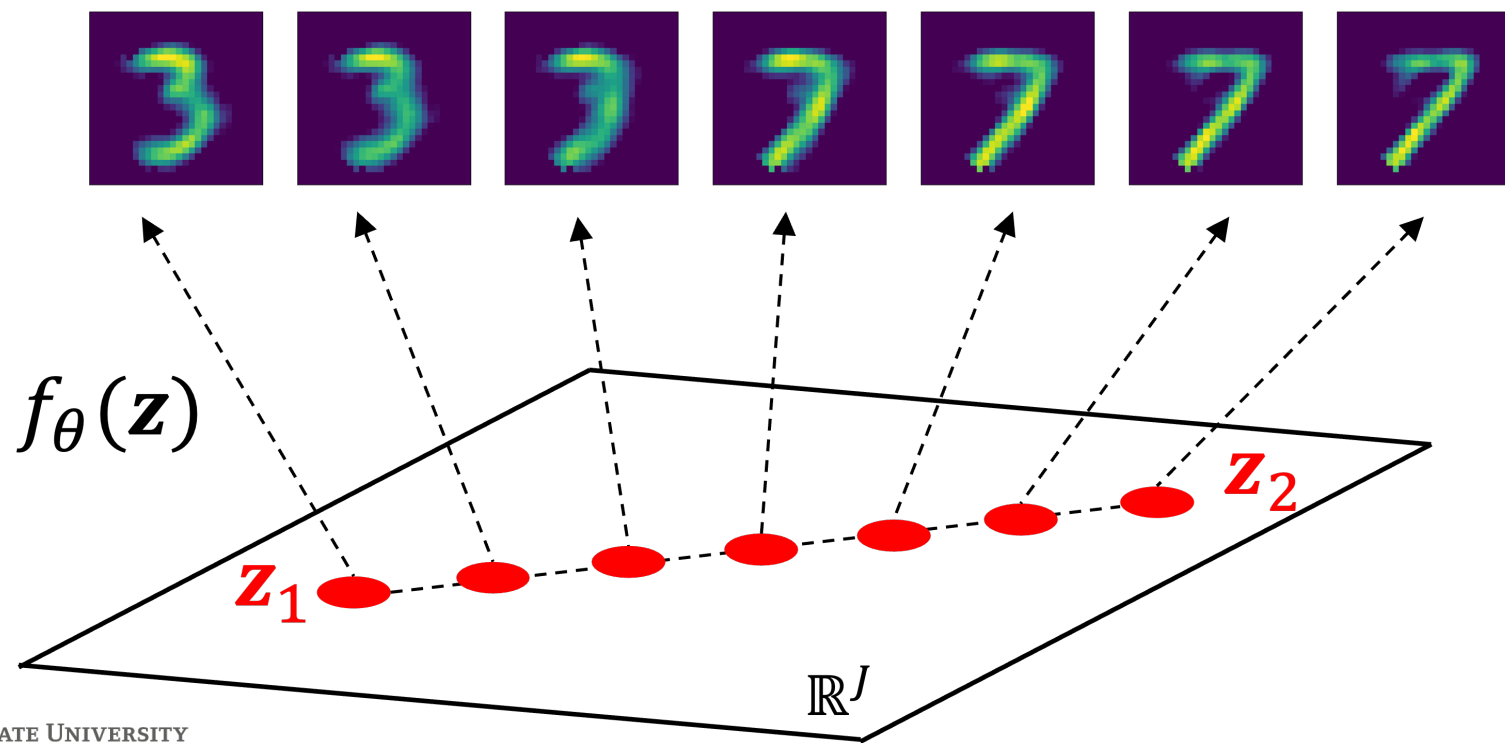
Reconstruction Example



Exploring the Latent Space...



Exploring the Latent Space... (cont.)



Simulation

- Check out

https://colab.research.google.com/drive/1tz_aNMLJjDqOtfK7MdGgylapDwf5FC-T?usp=sharing

Agenda

1. What are generative models? Examples?
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3. **How to evaluate generative models for images?**

Typical strategy for discriminative models

1. Obtain a test dataset with data $x_1, \dots, x_N$ and labels $y_1, \dots, y_N$
2. Evaluate the model for the estimated labels $\hat{y}_i = f_\theta(x_i)$
3. Compare $y_1, \dots, y_N$ and $\hat{y}_1, \dots, \hat{y}_N$ with respect to some distance metric
 - Binary/Categorical -> cross-entropy; Real-valued -> Avg Euclidean distance, etc.

Difficulty of Evaluating Generative Models

1. We can still obtain a test dataset with data $x_1, \dots, x_N$, but **no labels**
2. Even if we can generate $\tilde{x}_1, \dots, \tilde{x}_N$, but **no matching guarantee** that $x_i \approx \tilde{x}_i$
 - Attempt 1: Use human judgement
 - Humans are **costly**, and **varied**
 - No uniform criteria...
 - Attempt 2: Evaluate $\mathbb{E}_{X \sim p_{test}}[\log p_{\theta}(X)]$
 - Hard to compute in real-world...
 - Not robust to even tiny perturbations...

Evaluations for Images

- Candidate 1: Inception Scores
- Assumptions
 1. The generative model is trained on some **labelled** dataset
 2. We have a well-trained **classifier** $c(y|x)$
- Inception Score = Sharpness (S) * Diversity (D)
 - ...the higher the better

$$IS = S * D$$

Inception Score Example



Low sharpness

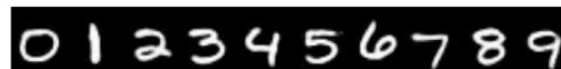


High sharpness

$$S = \exp \left(E_{\mathbf{x} \sim p} \left[\int c(y|\mathbf{x}) \log c(y|\mathbf{x}) dy \right] \right)$$



Low diversity



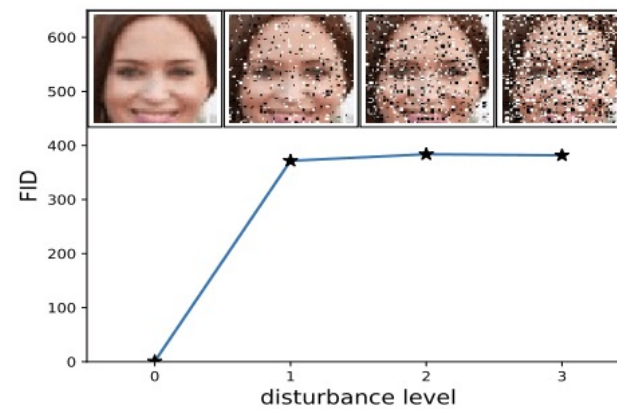
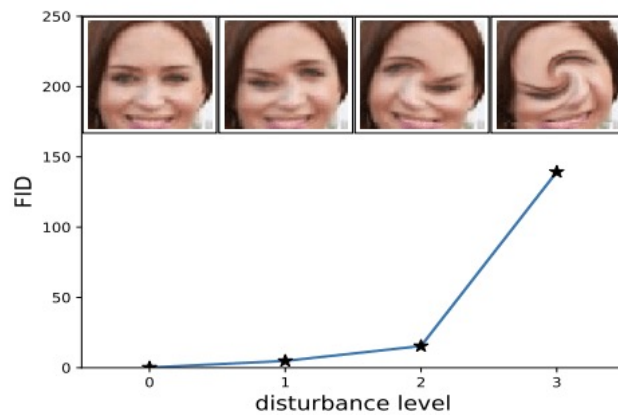
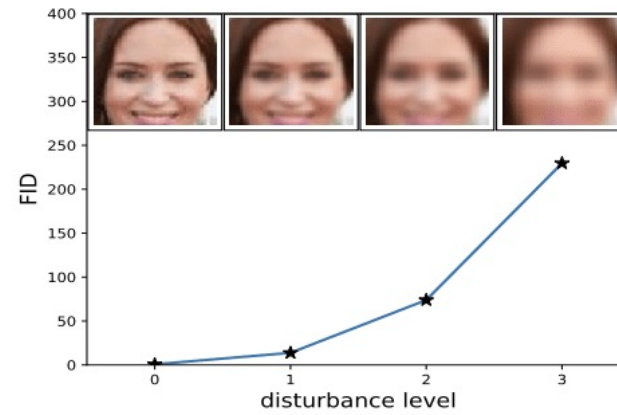
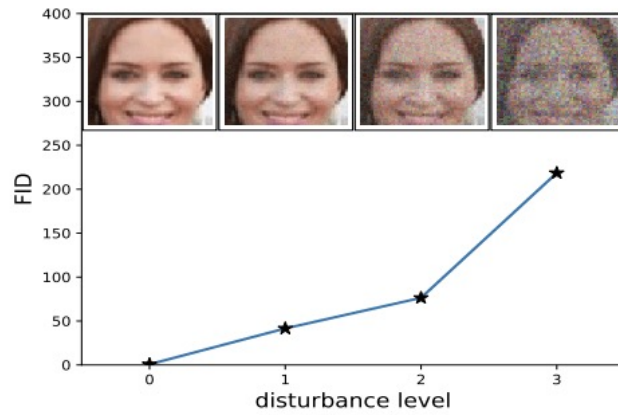
High diversity

$$D = \exp \left(-E_{\mathbf{x} \sim p} \left[\int c(y|\mathbf{x}) \log c(y) dy \right] \right)$$

Evaluations for Images (cont.)

- Candidate 2: Fréchet Inception Distance (FID)
 - ...the lower the better
- 1. Step 1: Let $\mathcal{G}$ denote the generated samples, and let $\mathcal{T}$ denote the test dataset
- 2. Step 2: Compute the feature vectors $\mathbf{F}_{\mathcal{G}}$ and $\mathbf{F}_{\mathcal{T}}$ (usually the last pooling layer of Inception v3, which is 2048-dim)
- 3. Step 3: Fit a multivariate Gaussian for each as $(\mu_{\mathcal{G}}, \Sigma_{\mathcal{G}})$ and $(\mu_{\mathcal{T}}, \Sigma_{\mathcal{T}})$
- 4. Step 4: The FID is equal to **Wasserstein-2** distance between them

FID



Homework

Choose one of the models below, change at least 3 parameters of the model; examine any difference in the generation quality (quantitatively or qualitatively)

1. VAE
2. DCGAN
3. Diffusion Model

- Send your report to liang.1439@osu.edu

Questions?

References

- S. Ermon, Stanford CS 236 Course, URL: deepgenerativemodels.github.io
- D. P. Kingma and M. Welling, An Introduction to Variational Autoencoder, <https://arxiv.org/pdf/1906.02691>
- M. N. Bernstein, Variational autoencoders, URL: <https://mbernste.github.io/posts/vae/>